Problem 1

(a)

$f_1(H) = P(H) \rightarrow$ Scope: $\{H\}$

H	$f_1(H)$
0	0.6
1	0.4

$f_2(H, V) = P(V|H) \rightarrow$ Scope: $\{H, V\}$

H	V	$f_2(H, V)$
0	0	0.8
0	1	0.2
1	0	0.3
1	1	0.7

$f_3(V, F) = P(F|V) \rightarrow$ Scope: $\{V, F\}$

V	F	$f_3(V, F)$
0	0	0.9
0	1	0.1
1	0	0.2
1	1	0.8

$f_4(H, V) = P(A = 1|H, V) \rightarrow$ Scope: $\{H, V\}$

H	V	$f_4(H, V)$
0	0	0.05
0	1	0.50
1	0	0.60
1	1	0.90

(b)

【Join】$f_{124}(H,V) = f_1(H)f_2(H,V)f_4(H,V) = P(A=1,H,V) \rightarrow$ Scope：$\{H,V\}$

H	V	$f_{124}(H,V)$
0	0	$0.8 \times 0.6 \times 0.05 = 0.024$
0	1	$0.2 \times 0.6 \times 0.50 = 0.060$
1	0	$0.3 \times 0.4 \times 0.60 = 0.072$
1	1	$0.7 \times 0.4 \times 0.90 = 0.252$

【Eliminate】$f_5(V) = \sum_H f_{124}(H,V) = P(A=1,V) \rightarrow$ Scope：$\{V\}$

V	$f_5(V)$
0	$0.024 + 0.072 = 0.096$
1	$0.060 + 0.252 = 0.312$

【Join】$f_{35}(V,F) = f_3(V,F)f_5(V) = P(A=1,V,F) \rightarrow$ Scope：$\{V,F\}$

V	F	$f_{35}(V,F)$
0	0	$0.096 \times 0.9 = 0.0864$
0	1	$0.096 \times 0.1 = 0.0096$
1	0	$0.312 \times 0.2 = 0.0624$
1	1	$0.312 \times 0.8 = 0.2496$

【Eliminate】$f_6(F) = \sum_V f_{35}(V,F) = P(A=1,F) \rightarrow$ Scope：$\{F\}$

F	$f_6(F)$
0	$0.0864 + 0.0624 = 0.1488$
1	$0.0096 + 0.2496 = 0.2592$

(c)

$$\because Z = \sum_F f_6(F) = P(A=1) = 0.1488 + 0.2592 = 0.408$$

$$\therefore P(F=0 \mid A=1) = \frac{f_6(0)}{Z} = \frac{0.1488}{0.408} \approx 0.365$$

$$\therefore P(F=1 \mid A=1) = \frac{f_6(1)}{Z} = \frac{0.2592}{0.408} \approx 0.635$$

Problem 2

(a)

$$\text{Let } f_1(D) = P(D), \quad f_2(D,T) = P(T|D), \quad f_3(D,I) = P(I|D)$$

$$f_4(D,T,I) = f_1(D)f_2(D,T)f_3(D,I)$$
$$f_4(0,0,0) = f_1(0) \times f_2(0,0) \times f_3(0,0) = 0.75 \times 0.85 \times 0.80 = 0.510$$
$$f_4(0,0,1) = f_1(0) \times f_2(0,0) \times f_3(0,1) = 0.75 \times 0.85 \times 0.20 = 0.1275$$
$$f_4(0,1,0) = f_1(0) \times f_2(0,1) \times f_3(0,0) = 0.75 \times 0.15 \times 0.80 = 0.090$$
$$f_4(0,1,1) = f_1(0) \times f_2(0,1) \times f_3(0,1) = 0.75 \times 0.15 \times 0.20 = 0.0225$$
$$f_4(1,0,0) = f_1(1) \times f_2(1,0) \times f_3(1,0) = 0.25 \times 0.15 \times 0.20 = 0.0075$$
$$f_4(1,0,1) = f_1(1) \times f_2(1,0) \times f_3(1,1) = 0.25 \times 0.15 \times 0.80 = 0.030$$
$$f_4(1,1,0) = f_1(1) \times f_2(1,1) \times f_3(1,0) = 0.25 \times 0.85 \times 0.20 = 0.0425$$
$$f_4(1,1,1) = f_1(1) \times f_2(1,1) \times f_3(1,1) = 0.25 \times 0.85 \times 0.80 = 0.170$$

$$f_5(T,I) = \sum\nolimits_D f_4(D,T,I) = P(T,I)$$
$$f_5(1,0) = f_4(0,1,0) + f_4(1,1,0) = 0.090 + 0.0425 = 0.1325$$

$$P(D=1|T=1,I=0) = \frac{f_4(1,1,0)}{f_5(1,0)} = \frac{0.0425}{0.1325} \approx 0.321$$

(b)

$$f_5(1,1) = f_4(0,1,1) + f_4(1,1,1) = 0.0225 + 0.170 = 0.1925$$

$$f_6(D,I) = f_1(D)f_3(D,I) = P(D,I)$$
$$f_6(0,0) = f_1(0)f_3(0,0) = 0.75 \times 0.80 = 0.600$$
$$f_6(0,1) = f_1(0)f_3(0,1) = 0.75 \times 0.20 = 0.150$$
$$f_6(1,0) = f_1(1)f_3(1,0) = 0.25 \times 0.20 = 0.050$$
$$f_6(1,1) = f_1(1)f_3(1,1) = 0.25 \times 0.80 = 0.200$$

$$f_7(I) = \sum\nolimits_D f_6(D,I) = P(I)$$
$$f_7(1) = f_6(0,1) + f_6(1,1) = 0.150 + 0.200 = 0.350$$

$$P(T=1|I=1) = \frac{f_5(1,1)}{f_7(1)} = \frac{0.1925}{0.350} = 0.550$$

Problem 3

(a)

Let $f_1(H) = P(H), \quad f_2(H,B) = P(B|H), \quad f_3(H,U) = P(U|H), \quad f_4(B,U,N) = P(N|B,U)$

$f_5(H,B,U) = f_1(H)f_2(H,B)f_3(H,U) = P(H,B,U)$
$f_5(0,0,0) = f_1(0)f_2(0,0)f_3(0,0) = 0.4 \times 0.8 \times 0.75 = 0.24$
$f_5(0,0,1) = f_1(0)f_2(0,0)f_3(0,1) = 0.4 \times 0.8 \times 0.25 = 0.08$
$f_5(0,1,0) = f_1(0)f_2(0,1)f_3(0,0) = 0.4 \times 0.2 \times 0.75 = 0.06$
$f_5(0,1,1) = f_1(0)f_2(0,1)f_3(0,1) = 0.4 \times 0.2 \times 0.25 = 0.02$
$f_5(1,0,0) = f_1(1)f_2(1,0)f_3(1,0) = 0.6 \times 0.4 \times 0.25 = 0.06$
$f_5(1,0,1) = f_1(1)f_2(1,0)f_3(1,1) = 0.6 \times 0.4 \times 0.75 = 0.18$
$f_5(1,1,0) = f_1(1)f_2(1,1)f_3(1,0) = 0.6 \times 0.6 \times 0.25 = 0.09$
$f_5(1,1,1) = f_1(1)f_2(1,1)f_3(1,1) = 0.6 \times 0.6 \times 0.75 = 0.27$

$f_6(B,U) = \sum_H f_5(H,B,U) = P(B,U)$
$f_6(0,0) = f_5(0,0,0) + f_5(1,0,0) = 0.24 + 0.06 = 0.30$
$f_6(0,1) = f_5(0,0,1) + f_5(1,0,1) = 0.08 + 0.18 = 0.26$
$f_6(1,0) = f_5(0,1,0) + f_5(1,1,0) = 0.06 + 0.09 = 0.15$
$f_6(1,1) = f_5(0,1,1) + f_5(1,1,1) = 0.02 + 0.27 = 0.29$

$f_7(B,U,N) = f_4(B,U,N)f_6(B,U) = P(B,U,N)$
$f_7(0,0,0) = f_4(0,0,0)f_6(0,0) = 1.00 \times 0.30 = 0.30$
$f_7(0,0,1) = f_4(0,0,1)f_6(0,0) = 0.00 \times 0.30 = 0.00$
$f_7(0,1,0) = f_4(0,1,0)f_6(0,1) = 0.20 \times 0.26 = 0.052$
$f_7(0,1,1) = f_4(0,1,1)f_6(0,1) = 0.80 \times 0.26 = 0.208$
$f_7(1,0,0) = f_4(1,0,0)f_6(1,0) = 0.30 \times 0.15 = 0.045$
$f_7(1,0,1) = f_4(1,0,1)f_6(1,0) = 0.70 \times 0.15 = 0.105$
$f_7(1,1,0) = f_4(1,1,0)f_6(1,1) = 0.05 \times 0.29 = 0.0145$
$f_7(1,1,1) = f_4(1,1,1)f_6(1,1) = 0.95 \times 0.29 = 0.2755$

$f_8(N) = \sum_{B,U} f_7(B,U,N) = P(N)$
$f_8(0) = f_7(0,0,0) + f_7(0,1,0) + f_7(1,0,0) + f_7(1,1,0) = 0.30 + 0.052 + 0.045 + 0.0145 = 0.4115$
$f_8(1) = f_7(0,0,1) + f_7(0,1,1) + f_7(1,0,1) + f_7(1,1,1) = 0.00 + 0.208 + 0.105 + 0.2755 = 0.5885$

(b)

Because rejection sampling only accepts samples that satisfy $N = 1$, so the acceptance probability is equal to $P(N = 1) \approx 0.589$.

(c)

$$\begin{aligned}\hat{P}(B = 1 \mid N = 1) &= \frac{\text{Accepted samples with } B = 1}{\text{Total accepted samples}} \\ &= \frac{5}{12} \approx 0.417\end{aligned}$$

Problem 4

(a)

$$\begin{aligned}\widehat{P}(B = 1|H = 1, N = 1) &= \frac{\sum_i w_i \mathbf{1}[B_i = 1]}{\sum_i w_i} \\ &= \frac{w_1 + w_3 + w_5}{w_1 + w_2 + w_3 + w_4 + w_5} \\ &= \frac{0.420 + 0.570 + 0.570}{0.420 + 0.480 + 0.570 + 0.000 + 0.570} \\ &= \frac{1.560}{2.040} \approx 0.765\end{aligned}$$

(b)

$$\begin{aligned}ESS &= \frac{(\sum_i w_i)^2}{\sum_i w_i^2} \\ &= \frac{(0.420 + 0.480 + 0.570 + 0.000 + 0.570)^2}{0.420^2 + 0.480^2 + 0.570^2 + 0.000^2 + 0.570^2} \\ &= \frac{(2.040)^2}{0.1764 + 0.2304 + 0.3249 + 0.0000 + 0.3249} \\ &= \frac{4.1616}{1.0566} \approx 3.94\end{aligned}$$

Problem 5

(a)

$$\begin{aligned}&\text{Let } h(B, H, U, N) = P(B|H)P(N|B, U) \\ &\because P(B \mid H, U, N) \propto P(B \mid H)P(N \mid B, U) \\ &\therefore h(1, 0, 1, 1) = P(B = 1|H = 0)P(N = 1|B = 1, U = 1) = 0.2 \times 0.95 = 0.19 \\ &\therefore h(0, 0, 1, 1) = P(B = 0|H = 0)P(N = 1|B = 0, U = 1) = 0.8 \times 0.80 = 0.64 \\ &\therefore P(B = 1|H = 0, U = 1, N = 1) = \frac{h(1, 0, 1, 1)}{h(1, 0, 1, 1) + h(0, 0, 1, 1)} = \frac{0.19}{0.19 + 0.64} \approx 0.229\end{aligned}$$

(b)

In Gibbs Sampling, B is resampled from its conditional distribution given the current values of H, U, and N. Therefore, the probability that B flips from 0 to 1 is the same as the probability calculated in part (a), which is 0.229.

(c)

From above:

$$f_6(B,U) = \sum_H f_5(H,B,U) = P(B,U)$$

$$f_6(0,0) = 0.30 \quad f_6(0,1) = 0.26$$

$$f_6(1,0) = 0.15 \quad f_6(1,1) = 0.29$$

Then:

$$P(B=0,U=1) = 0.26 \quad P(B=1,U=1) = 0.29$$

$$P(B=0|U=1) = \frac{0.26}{0.26+0.29} \approx 0.473$$

$$P(B=1|U=1) = \frac{0.29}{0.26+0.29} \approx 0.527$$

Therefore:

$$P(B=1 \mid U=1) > P(B=0 \mid U=1)$$

Problem 6

(a)

$$P(X_1 = M|X_0 = I) = 0.2$$

(b)

$$\begin{aligned} P(X_2 = M|X_0 = I) &= P(X_1 = I|X_0 = I)P(X_2 = M|X_1 = I) \\ &\quad + P(X_1 = B|X_0 = I)P(X_2 = M|X_1 = B) \\ &\quad + P(X_1 = M|X_0 = I)P(X_2 = M|X_1 = M) \\ &= 0.5 \times 0.2 + 0.3 \times 0.2 + 0.2 \times 0.7 \\ &= 0.10 + 0.06 + 0.14 \\ &= 0.300 \end{aligned}$$

(c)

Let the state order be (I, B, M)

$$P = \begin{pmatrix} 0.5 & 0.3 & 0.2 \\ 0.2 & 0.6 & 0.2 \\ 0.1 & 0.2 & 0.7 \end{pmatrix} \quad P^2 = \begin{pmatrix} 0.330 & 0.370 & 0.300 \\ 0.240 & 0.460 & 0.300 \\ 0.160 & 0.290 & 0.550 \end{pmatrix} \quad P^3 = \begin{pmatrix} 0.269 & 0.381 & 0.350 \\ 0.242 & 0.408 & 0.350 \\ 0.193 & 0.332 & 0.475 \end{pmatrix}$$

Therefore $P(X_3 = B|X_0 = I) = 0.381$